DTC P0087:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".

Details On DTCs

DESCRIPTION	Fuel pressure sensor circuit range/performance problem	
DETECTION CONDITION	Determination conditions	• The actual fuel pressure is lower than the target fuel pressure by **1.2 MPa {12 kgf/cm^2, 174 psi}** or more even though the fuel pressure feedback amount is maintained high.
	Preconditions	• Engine speed: **500 rpm or more** *1 • The following DTCs are not detected: — Fuel pressure sensor: P0089:00, P0192:00, P0193:00 — High pressure fuel pump: P0091:00, P0092:00 *1: Value can be verified by displaying PIDs using M-MDS
	Malfunction determination period	• **5 s** period
	Drive cycle	• 1
	Self test type	• CMDTC self test
	Sensor used	• Fuel pressure sensor
FAIL-SAFE FUNCTION	• Limits intake air amount • Stops high pressure fuel pump control	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Illuminates check engine light.	
POSSIBLE CAUSE	• Air suction in fuel line due to fuel is runout • High pressure fuel pump connector or terminals malfunction • Fuel pressure sensor connector or terminals malfunction • PCM connector or terminals malfunction • Fuel pressure sensor malfunction • Fuel delivery system malfunction (insufficient fuel pressure) • Fuel injector malfunction • High pressure fuel pump malfunction — Relief valve (built-into high pressure fuel pump) malfunction • PCM malfunction	

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The PCM calculates the target fuel pressure appropriate to the engine conditions relative to the actual fuel pressure based on the fuel pressure sensor signal and provides feedback to the high pressure fuel pump control. If the difference between the actual fuel pressure and the target fuel pressure relative to the feedback amount is large, the PCM determines a malfunction in the fuel pressure sensor and stores a DTC.

Repeatability Verification Procedure
1. Warm up the engine to allow the engine coolant temperature to reach **80 °C {176 °F} or more**.
2. Start the engine and leave it idling for **1 min**.

   Note
   - The target calculated by the PCM is increased and the DTC determination conditions are met by repeating the Step 3 operation.

3. Shift to 1st gear and accelerate/decelerate the vehicle to **30—50 km/h** {**19—31 mph**}.

   Note
   - Match the engine coolant temperature in the recorded FREEZE FRAME DATA/snapshot data, the vehicle speed, and engine speed values to the best extent possible while driving the vehicle.

4. Try to reproduce the malfunction by driving the vehicle for 5 min based on the values in the FREEZE FRAME DATA/snapshot data.

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
FUEL_PRES	Fuel pressure input from fuel pressure sensor	KPa {MPa}, mBar {Bar}, psi, in H20	• Displays fuel pressure
	Fuel pressure sensor voltage	V	**Ignition switched ON (engine off)** • Fuel pressure is approx. 16.91 MPa {172.4 kgf/cm^2, 2453 psi}: Approx. 2.91 V **Idle (after warm up)** • Fuel pressure is 3.44—3.95 MPa {35.1—40.2 kgf/cm^2, 499—572 psi}: 0.98—1.05 V (ECT is 90 °C {194 °F})

Function Inspection Using M-MDS

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P0087:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3	**PURPOSE: RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFICATION IF MALFUNCTION CAUSED BY LACK OF FUEL** • Verify the snapshot data FLI. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the snapshot data FLI **5% or less**?	Yes	Refill the fuel. Go to the troubleshooting procedure to perform the procedure from Step 3.
		No	Go to the next step.
5	**PURPOSE: VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
6	**PURPOSE: VERIFY CONNECTOR CONNECTIONS** • Start the engine. • Access the FUEL_PRES PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Does the PID value fluctuate when the following connectors are shaken? — Fuel pressure sensor — PCM	Yes	Repair or replace the applicable wiring harness or connector parts. Go to the troubleshooting procedure to perform the procedure from Step 3.
		No	Go to the next step.
7	**PURPOSE: VERIFY FUEL PRESSURE (LOW-SIDE) MALFUNCTION** **Note** • Verify the fuel pressure on the low pressure side with the operation of the high pressure fuel pump turned off. • Bleed the remaining pressure in the fuel line using the following procedure. 1. Switch the ignition off. 2. Disconnect the high pressure fuel pump connector. 3. Disconnect the fuel pump relay. 4. Start the engine and wait until the engine stalls. • Switch the ignition off. • Install the fuel pump relay. • Switch the ignition ON (engine off). • Display PID FUEL_PRES and simulation item FP using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Turn simulation item FP on. • Is the FUEL_PRES PID value **430 kPa {4.38 kgf/cm^2, 62.4 psi} or more**?	Yes	Go to the troubleshooting procedure to perform the procedure from Step 2.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1
  - — Inspect the fuel supply system (low pressure side).
- Step 2
  - — Perform a unit inspection of the fuel pressure sensor.
- Step 3—4
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY IF MALFUNCTION RELATED TO FUEL PIPE (LOW PRESSURE SIDE) AFFECTS DIAGNOSTIC RESULTS** • Inspect the fuel pump unit. (See FUEL PUMP UNIT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the fuel pump unit, then go to Step 3. (See FUEL PUMP UNIT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Inspect the fuel system pipes (low to high pressure sides) for fuel leakage and restriction. • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results, then go to Step 3. • If there is no malfunction: — Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
2	**PURPOSE: DETERMINE INTEGRITY OF FUEL PRESSURE SENSOR** • Inspect the fuel pressure sensor. (See FUEL PRESSURE SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the fuel distributor, then go to the next step. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
3	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
4	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.